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United States Patent	12414875
Kind Code	B1
Date of Patent	September 16, 2025
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Anterior vitrectomy apparatus

Abstract

A portable, compact, standalone anterior vitrectomy apparatus includes a housing sized and configured to fit in the palm of a human hand. A vitrectomy cutter connectable to an air tubing connector in the housing via an air tubing. A pressure reservoir in the housing communicates with an air control valve which in turn communicates with the air tubing connector. An internal air compressor, arranged inside the housing, communicates with an air regulator for supplying compressed air to the pressure reservoir. A speed control switch controllably interface with the air control valve. A processor controllably interface with the speed control switch and control the air control valve through the speed control switch such that the air control valve delivers pulses of air from the pressure reservoir through the air control valve to drive the vitrectomy cutter.

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Family ID:	1000007516911
Appl. No.:	18/517679
Filed:	November 22, 2023

Related U.S. Application Data

continuation-in-part parent-doc US 16859250 20200427 ABANDONED child-doc US 18517679
us-provisional-application US 62838472 20190425

Publication Classification

Int. Cl.: A61F9/007 (20060101); A61B90/30 (20160101); A61B17/00 (20060101)

U.S. Cl.:

CPC **A61F9/00754** (20130101); **A61B90/30** (20160201); A61B2017/00424 (20130101);
A61B2017/00544 (20130101); A61B2017/00973 (20130101); A61B2090/306
(20160201); A61B2217/005 (20130101); A61B2217/007 (20130101)

Field of Classification Search

CPC: A61F (9/00754); A61F (9/00736)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a Continuation-in-Part of U.S. patent application Ser. No. 16/859,250, filed on Apr. 27, 2020, which in turn claims the benefit of U.S. Provisional Patent Application No. 62/838,472, filed on Apr. 25, 2019, which are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to medical devices, and more particularly, to a portable, compact, standalone anterior vitrectomy apparatus which can be used to cut vitreous humor from a subject's eye.

BACKGROUND OF THE INVENTION

(2) The human eye is filled with a gel-type substance known as vitreous gel, or vitreous humor. Vitreous humor is a thick, colorless, gel-like fluid which fills the interior space of the eyeball between the lens and the retina and helps the eyeball maintain its generally spherical shape.

(3) In some cases, it may be necessary or desirable to remove the vitreous humor from an eye. Such a procedure is known as a vitrectomy. Vitrectomy may facilitate access to repair or prevent retinal detachment, retinal tears, or severe proliferative retinopathy. Vitrectomy may also reduce loss of vision in the case of vitreous hemorrhage. In cataract surgery, any vitreous that migrates forward towards the front of the eye must be cut and removed to prevent vision loss.

(4) Various types of vitrectomy devices are known in the art. These devices have different designs with various features. However, all vitrectomy devices have the same basic functional mechanism. These devices cut the vitreous humor out of the eye using a reciprocating cutting blade (driven either pneumatically or electrically), combined with a source of vacuum that removes the vitreous humor and irrigation which replaces the removed mass with fluid.

(5) Conventional vitrectomy devices typically fall into two categories: posterior vitrectomy apparatus and anterior vitrectomy apparatus. Posterior vitrectomy apparatus are typically used for retinal and other surgeries which are performed at the back of the eye. These surgeries may be complex and may need to be combined with other retinal repairing tools such as fiber optic light

sources, pneumatic scissors, etc. Anterior vitrectomy apparatus are typically simpler than posterior vitrectomy apparatus and may be used in cataract and other surgical procedures which are performed at the front of the eye. Although various standalone posterior vitrectomy apparatus exist, there exist very few standalone anterior vitrectomy apparatus. These anterior vitrectomy apparatus may exhibit poor performance specs (**800** cuts per minute or less, absent basic fluidics control). Most anterior vitrectomy drivers are integrated into large cataract apparatus, as they are usually used in conjunction. As a result, these apparatus may be large and costly.

(6) However, there is a demonstrated global need, particularly in rural settings, for an ultraportable, high-performance, standalone anterior vitrectomy apparatus. Many places which do not have access to cataract apparatus may require that the surgeon use a standalone anterior vitrectomy apparatus which he or she can easily carry or transport.

(7) Accordingly, there is an established need for a portable, compact, standalone anterior vitrectomy apparatus which can be used to cut vitreous humor from a subject's eye.

SUMMARY OF THE INVENTION

(8) The present invention is directed to a portable, compact, standalone anterior vitrectomy apparatus which can be used to cut vitreous humor from a subject's eye. The anterior vitrectomy apparatus may include a housing which contains or encloses at least some of the functional components of the apparatus. The housing may be sized and configured to fit in the palm of a human hand.

(9) In a first implementation, an anterior vitrectomy apparatus may include: an air compressor, configured to generate compressed air; a pressure reservoir, configured to store compressed air received from the air compressor; a regulator, configured to vary a delivery of compressed air from the air compressor to the pressure reservoir to adjust air pressure within the pressure reservoir; a speed control switch; an air control valve, configured to receive compressed air from the pressure reservoir and to produce pulses of compressed air in accordance with a selective adjustment of the speed control switch; and an air tubing connector, configured to receive the pulses of compressed air from the air control valve. The anterior vitrectomy apparatus may further include an air tubing, connectable to the air tubing connector, and a handheld vitrectomy cutter, connectable to the air tubing to receive the pulses of compressed air from the air control valve via the air tubing connector and the air tubing. A processor may controllably interface with the speed control switch. The processor may include a memory storing software instructions configured to cause the processor to execute the operation of producing an operation of the air control valve through the speed control switch such that the air control valve delivers said pulses of compressed air to the air tubing connector to drive the handheld vitrectomy cutter. The anterior vitrectomy apparatus may further include a housing containing the air compressor, the pressure reservoir, the regulator, the air control valve, and the processor, the housing sized to fit in and be held by a human hand. The air tubing connector and the speed control switch may be arranged on an exterior of the housing.

(10) In a second aspect, the anterior vitrectomy apparatus may further include a release valve, contained inside the housing. The release valve may be configured to selectively exhaust compressed air from between the compressor and the regulator. The processor may controllably interface with the release valve and may be configured to cause the release valve to exhaust compressed air from between the air compressor and the regulator in the absence of said operation of the air control valve by the processor.

(11) In another aspect, the anterior vitrectomy apparatus may further include a pressure sensor, contained inside the housing. The pressure sensor may be configured to sense air pressure between the air compressor and the regulator. The processor may controllably interface with the pressure sensor and may be configured to cause the release valve to exhaust compressed air from between the air compressor and the regulator when the processor detects a pressure sensing by the pressure sensor between the air compressor and the regulator greater than a predetermined threshold, in the absence of said operation of the air control valve by the processor.

- (12) In another aspect, the predetermined threshold may be about zero.
- (13) In another aspect, the processor may be configured to cause the release valve not to exhaust compressed air from between the air compressor and the regulator during said operation of the air control valve by the processor.
- (14) In yet another aspect, the anterior vitrectomy apparatus may further include a pressure gauge inside the housing. The pressure gauge may interface with the processor and the pressure reservoir and may be configured to indicate pressure of air in the pressure reservoir to the processor.
- (15) In another aspect, the anterior vitrectomy apparatus may further include a syringe connectable to the handheld vitrectomy cutter to suction and collect vitreous humor from the handheld vitrectomy cutter.
- (16) In another aspect, the anterior vitrectomy apparatus may further include a foot pedal connector on the exterior of the housing. The foot pedal connector may be configured for the coupling thereto of a foot pedal external to the housing such that the foot pedal connector connects the foot pedal to the processor to facilitate said operation of the air control valve.
- (17) In another aspect, the speed control switch may include a rotatable dial.
- (18) In yet another aspect, the anterior vitrectomy apparatus may further include a reset button on housing. The reset button may be configured to facilitate a resetting of the processor.
- (19) In another aspect, the anterior vitrectomy apparatus may further include a peristaltic pump and a vacuum line. The peristaltic pump may be located inside the housing. The vacuum line may provide fluid communication between the peristaltic pump and the handheld vitrectomy cutter to suction and collect vitreous humor from the handheld vitrectomy cutter.
- (20) In another aspect, the peristaltic pump may be driven by a stepper motor driver located inside the housing and interfacing with the processor.
- (21) In another aspect, the anterior vitrectomy apparatus may further include a vacuum sensor in fluid communication with the vacuum line and configured to provide a measurement of air vacuum pressure in the vacuum line to the processor.
- (22) In yet another aspect, the anterior vitrectomy apparatus may further include an air pump and air tubing. The air pump may be located inside the housing and may be configured to provide pressurized air. The air tubing may provide fluid communication between the air pump and an infusion line external to the housing and may be configured to be inserted into a subject's eye.
- (23) In another aspect, the anterior vitrectomy apparatus may further include a pinch valve, a pinch valve tube and an infusion line. The pinch valve may be located inside the housing. The infusion line may be configured to be inserted into a subject's eye. The pinch valve tube may provide fluid communication between a source of irrigation fluid and the infusion line through the pinch valve.
- (24) In another aspect, the source of irrigation fluid may be configured to feed irrigation fluid to the pinch valve by gravity.
- (25) In another aspect, the anterior vitrectomy apparatus may further include an air pump and air tubing. The air pump may be located inside the housing and may be configured to provide pressurized air. The air tubing may provide fluid communication between the air pump and the infusion line.
- (26) In yet another aspect, the anterior vitrectomy apparatus may further include a light source and a fiber-optic cable. The light source may be arranged inside the housing. The fiber-optic cable may be configured to transport light emitted by the light source to a fiber-optic handpiece external to the housing and configured to be inserted into a subject's eye.
- (27) These and other objects, features, and advantages of the present invention will become more readily apparent from the attached drawings and the detailed description of the preferred embodiments, which follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The preferred embodiments of the invention will hereinafter be described in conjunction with the appended drawings provided to illustrate and not to limit the invention, where like designations denote like elements, and in which:
- (2) FIG. 1 presents a top front isometric view of an anterior vitrectomy apparatus in accordance with a first illustrative embodiment of the present invention;
- (3) FIG. 2 presents a top rear isometric view of the anterior vitrectomy apparatus of FIG. 1;
- (4) FIG. 3 presents a left side elevation view of the anterior vitrectomy apparatus of FIG. 1;
- (5) FIG. 4 presents a functional block diagram of the anterior vitrectomy apparatus of FIG. 1;
- (6) FIG. 5 presents a side view of the vitrectomy cutter of the anterior vitrectomy apparatus of FIG. 1, more particularly illustrating use of the vitrectomy cutter to remove vitreous humor from the eye of a patient in typical application of the anterior vitrectomy apparatus;
- (7) FIG. 6 presents a top front isometric view of an anterior vitrectomy apparatus in accordance with a second illustrative embodiment of the present invention;
- (8) FIG. 7 presents a top rear isometric view of the anterior vitrectomy apparatus of FIG. 6;
- (9) FIG. 8 presents a functional block diagram of the anterior vitrectomy apparatus of FIG. 6;
- (10) FIG. 9 presents a top front isometric view of an anterior vitrectomy apparatus in accordance with a third illustrative embodiment of the present invention;
- (11) FIG. 10 presents a functional block diagram of the anterior vitrectomy apparatus of FIG. 9; and
- (12) FIG. 11 presents a functional block diagram of an anterior vitrectomy apparatus in accordance with a third illustrative embodiment of the present invention, comprising an internal air compressor, pressure sensor and release valve.
- (13) Like reference numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(14) The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments or the application and uses of the described embodiments. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description herein, the terms “upper”, “lower”, “left”, “rear”, “right”, “front”, “vertical”, “horizontal”, and derivatives thereof shall relate to the invention as oriented in FIG. 1. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(15) Shown throughout the figures, the present invention is directed toward a portable, compact, standalone anterior vitrectomy apparatus which can be used to cut vitreous humor from a subject's eye.

(16) Referring initially to FIGS. 1-5, a first illustrative embodiment of the anterior vitrectomy apparatus is generally indicated by reference numeral **100**. The anterior vitrectomy apparatus **100** may include a housing **102** which contains or encloses at least some of the functional components of the anterior vitrectomy apparatus **100**. The housing **102** may have a size and configuration which

render the housing **102** amenable to being comfortably and securely held in the palm of one hand. In some embodiments, the housing **102** may include a housing bottom panel **104** and a housing top panel **112** disposed in parallel, spaced-apart relationship to each other. A pair of parallel, spaced-apart housing side panels **106** may extend between the housing bottom panel **104** and the housing top panel **112** at opposite left and right sides of the housing **106**, as shown. A housing front panel **108** and a housing rear panel **110** may extend between the housing bottom panel **104** and the housing top panel **112** and between the housing side panels **106**.

(17) As shown in FIG. 1, an air tubing connector **126** may be provided on the exterior of the housing **102**, such as on the housing front panel **108**, as shown. Air tubing **122** may be connectable to the air tubing connector **126**. A handheld vitrectomy cutter **116** may be connectable to the air tubing **122**. In some embodiments, the vitrectomy cutter **116** may have a conventional design known by those skilled in the art. The vitrectomy cutter **116** may include a handpiece **118** which is connectable to the air tubing **122**. An air-driven oscillation motor (not illustrated) may be provided in the handpiece **118**. A reciprocating cutting blade **120** may extend from the handpiece **118**. The cutting blade **120** may be operably engaged by the oscillation motor to reciprocate the oscillation motor responsive to air actuation.

(18) As illustrated in FIG. 4, in some embodiments, a syringe **154** may be connectable to the vitrectomy cutter **116** through a syringe tubing **124**. In typical application of the anterior vitrectomy apparatus **100**, which will be hereinafter described, the syringe **154** may be operated to create a vacuum or suction on the syringe tubing **124** in order to draw vitreous humor to the syringe, and thereby facilitate collection of vitreous humor removed from the eye **170** of a patient **168** through the cutting blade **120** of the vitrectomy cutter **116**, as illustrated in FIG. 5.

(19) As illustrated in FIG. 4, an air control valve **130**, pressure reservoir **132** and regulator **150** may be provided in the housing **102**. The air control valve **130** may communicate with the vitrectomy cutter **116** through the air tubing **122**. In turn, the pressure reservoir **132** may communicate with the air control valve **130**. The regulator **150** may communicate with the pressure reservoir **132** and may be configured to regulate the pressure of air provided to the pressure reservoir **132** so that the pressure of air within the pressure reservoir **132** is maintained at a desired level (e.g., 30 PSI).

(20) As illustrated in FIG. 2, an air compressor connector **160** may be provided on the exterior of the housing **102**. The air compressor connector **160** may communicate with the regulator **150** in the housing **102**. The air compressor connector **160** may be configured for connection to an air compressor **148** (FIG. 4) outside the housing **102** in operation of the anterior vitrectomy apparatus **100**, which will be hereinafter described. For instance and without limitation, the air compressor **148** may be a conventional type of compressor which is used to supply a source of compressed air in vitrectomy procedures. In some embodiments, the air compressor connector **160** may be a quick-disconnect connector known by those skilled in the art.

(21) As illustrated in FIGS. 1 and 3, a speed control switch **142** may be provided on the exterior of the housing **102**. The illustration of FIG. 4, which shows user-operable and/or user interface components in broken lines, shows that the speed control switch **142** may controllably interface with the air control valve **130**. The speed control switch **142** may enable the operator of the anterior vitrectomy apparatus **100** to adjust the reciprocating speed of the cutting blade **120** of the vitrectomy cutter **116** appropriately. In some embodiments, the speed control switch **142** may include a rotatable dial which is selectively positional to various CPM (cuts per minute) values such as 100, 600, 1200 and 2000, for example and without limitation.

(22) As further shown in FIG. 4, a pressure gauge **134** may be provided in the housing **102**. The pressure gauge **134** may communicate with the pressure reservoir **132** to measure the pressure of air therein. The pressure gauge **134** may be configured to indicate the pressure of air in the pressure reservoir **132**. A microprocessor **136** or other processing unit (which may include a memory storing software instructions to be executed by the microprocessor **136** or other processing unit) may controllably interface with the speed control switch **142** and may interface with the pressure gauge

134. The microprocessor **136** may be programmed to control the air control valve **130** through the speed control switch **142** such that the air control valve **130** delivers pulses of air from the pressure reservoir **132** through the air control valve **130** to drive the vitrectomy cutter **116**.

(23) As illustrated in FIG. 2, a power input port **164** may be provided on the exterior of the housing **102**, such as on the housing rear panel **110**. The power input port **164** may be connected to the microprocessor **136**. The power input port **164** may be configured to receive a power cord (not illustrated) which may be connectable to a standard 120- or 240-volt power outlet (not illustrated) or other applicable electrical power source.

(24) As further shown in FIGS. 1 and 4, in some embodiments, at least one status LED **138** may be provided on the exterior of the housing **102**. The status LED **138** may be configured to provide a visual indication informative of the operational status of the anterior vitrectomy apparatus **100**. For example, the status LED **138** may illuminate a first color, such as green, to indicate a “ready to operate” status (e.g., a status in which pressure of air indicated by the pressure gauge is within a first or normal range); a second color, such as red, to indicate a “excessively-high pressure” status (e.g., a status in which pressure of air indicated by the pressure gauge is higher than a first threshold, wherein the first threshold is greater than or equal to the first range); and a third color, such as blue, to indicate a “excessively-low pressure” status (e.g., a status in which pressure of air indicated by the pressure gauge is lower than a second threshold, wherein the second threshold is less than or equal to the first range). Alternative embodiments are contemplated in status LED may illuminate in different colors and/or illumination patterns (e.g., continuous, flashing, etc.) to provide additional feedback regarding operation or status of the apparatus.

(25) As illustrated in FIG. 4, in some embodiments, a foot pedal **140**, external to the housing **102**, may be controllably connectable to the microprocessor **136**, typically through a foot pedal connector **158** (FIG. 1). The foot pedal connector **158** can be provided on the housing front panel **108**, for instance and without limitation. The foot pedal **140** may facilitate foot operation of the vitrectomy cutter **116** through the air control valve **130** while leaving the user's hands free to operate the handpiece **118** and carry out other tasks. For example, the foot pedal **140** may be configured to selectively operate the air control valve **130** to enable or disable airflow from the pressure reservoir **132** to the vitrectomy cutter **116**, to activate or deactivate cutting, respectively.

(26) As illustrated in FIG. 2, in some embodiments, a reset button **162** may be provided on the exterior of the housing **102**, such as on the housing rear panel **110**. The reset button **162** may operably interface with the microprocessor **136** (FIG. 4) to facilitate resetting of the software run by the microprocessor **136**.

(27) In typical application of the anterior vitrectomy apparatus **100**, the housing **102** may be easily and securely carried or transported by an operator of the apparatus **100** to a surgery location. As illustrated in FIG. 5, the apparatus **100** may be used to cut and remove vitreous humor from the eye **170** of a subject or patient **168**. Accordingly, a power cord (not illustrated) may be inserted in the power input port **164** (FIG. 2) and connected to a standard 120 or 240-volt electrical outlet (not illustrated) or other applicable electrical power source. The air tubing **122** which extends from the vitrectomy cutter **116** may be connected to the air tubing connector **126** (FIG. 1) on the housing **102**. In some applications, the foot pedal **140** may be connected to the foot pedal connector **158** on the housing **102** to interconnect the foot pedal **140** to the microprocessor **136**. As illustrated in FIG. 4, in some applications, the syringe **154** may be connected to the vitrectomy cutter **116** typically through the syringe tubing **124**. The compressor **148** may be connected to the air compressor connector **160** (FIG. 2) on the housing **102**, thereby providing fluid communication between the compressor **148** and the regulator **150**.

(28) The compressor **148** may be switched on, and compressed air may flow from the compressor **148** through the regulator **150** to the pressure reservoir **132**. Responsive to input from the pressure gauge **134**, the microprocessor **136** may monitor the pressure of the air in the pressure reservoir **132** and energize or illuminate the status LED **138** accordingly. For example, the microprocessor

136 (FIG. 4) may initially energize and illuminate the status LED **138** in the color green, which may indicate the “ready to operate” status of the anterior vitrectomy apparatus **100**.

(29) By manipulation of the speed control switch **142** on the housing **102**, the operator of the anterior vitrectomy apparatus **100** may select the desired operational speed of the vitrectomy cutter **116**. The operator may then depress the foot pedal **140** to cause the microprocessor **136**, controlled by the supporting software, to open the air control valve **130** to a position and at a rate and timing which cause compressed air from the pressure reservoir **132** to flow in pulses of air that drive the vitrectomy cutter **116** through the air tubing **122**. The desired operational speed of the vitrectomy cutter **116** may be set using the speed control switch **142**.

(30) With reference to FIG. 5, the operator may grip the vitrectomy cutter **116** at the handpiece **118** and direct the tip of the cutting blade **120** through the front of the eye **170** of the patient **168** and into the vitreous humor in the interior of the eye **170**. At the vitrectomy cutter **116**, the compressed air may flow from the air tubing **122** into the handpiece **118** and the oscillation motor (not illustrated) inside the handpiece **118**. The oscillation motor may reciprocate the cutting blade **120** with respect to the handpiece **118**. Simultaneously, the syringe **154** (FIG. 4) may be operated to draw the vitreous humor from the patient's eye **170** through the syringe tubing **124** into the syringe **154**.

(31) After the desired quantity of vitreous humor has been removed from the patient's eye **170**, the operator may terminate operation of the vitrectomy cutter **116** by releasing the foot pedal **140**. This action may cause the microprocessor **136** to close the air control valve **130**, which may prevent further flow of the compressed air from the pressure reservoir **132** through the air control valve **130** to the vitrectomy cutter **116**. The operator may then remove the cutting blade **120** from the patient's eye **170** and complete the procedure.

(32) Throughout operation of the vitrectomy cutter **116**, the operator may continually receive input from the microprocessor **136**, via the status LED **138**, which indicates the pressure of air in the pressure reservoir **132**. For example and without limitation, in the event that the pressure of air in the pressure reservoir **132** exceeds a predetermined threshold pressure, the microprocessor **136** may illuminate the status LED **138** in the color red, which may indicate damage or malfunction of the unit. In the event that the pressure of air in the pressure reservoir **132** falls below the predetermined threshold pressure, the microprocessor **136** may illuminate the status LED **138** in the color blue, allowing the user to become aware of the low-pressure situation and act accordingly (e.g., by checking that the air compressor **148** is correctly connected to the compressor connector **160**, turned on, and set at or above the minimal pressure (e.g., 40 PSI)).

(33) After use of the anterior vitrectomy apparatus **100** is completed, the power cord (not illustrated) may be removed from the power input port **164** (FIG. 2) in the housing **102**. The air compressor **148** (FIG. 4) may be disconnected from the air compressor connector **160** on the housing **102**. The air tubing **122** may be disconnected from the air tubing connector **126** on the housing **102**. The foot pedal **140** may be disconnected from the foot pedal connector **158** (FIG. 1). The housing **102**, vitrectomy cutter **116**, air tubing **122**, air compressor **148** and other components used in operation of the anterior vitrectomy apparatus **100** may be easily stored and carried or transported for use in multiple locations. In some embodiments, the operational software for the microprocessor **136** may be reset by depression of the reset button **162**.

(34) Referring next to FIGS. 6-8, a second illustrative embodiment of the anterior vitrectomy apparatus of the present invention is generally indicated by reference numeral **200**. In the anterior vitrectomy apparatus **200**, elements which are analogous to the respective elements of the anterior vitrectomy apparatus **100** that was heretofore described with respect to FIGS. 1-5 are designated by the same respective numerals in the 200-299 series in FIGS. 6-8.

(35) Similarly to the previous embodiment, the anterior vitrectomy apparatus **200** of the present embodiment may include a microprocessor **236**, regulator **250**, pressure reservoir **232**, air control valve **230**, and pressure gauge **234** contained within a housing **202** (FIGS. 6 and 7). The housing

202 is compact, manually portable, and, in some embodiments, is small enough to fit in a single hand of a user. The housing **202** may be generally cubic or rectangular and may include housing bottom, side, front, rear and top panels **204**, **206**, **208**, **210** and **212**. Also similarly to the previous embodiment, as shown in FIGS. **6** and **7**, the anterior vitrectomy apparatus **200** may include an air tubing connector **226**, status LED **238**, speed control switch **242**, air compressor connector **260**, reset button **262**, power input port **264**, and foot pedal connector **258** provided on the housing **202**. As with the previous embodiment, a vitrectomy cutter **216** comprising a handpiece **218** and cutting blade **220** may be connected to the air tubing connector **226** via an air tubing **222**. An air compressor **248** (FIG. **8**) may be connected to the regulator **250** via the air compressor connector **260**.

(36) As shown in FIG. **8**, in order to collect vitreous humor removed from the eye of a patient, the present embodiment includes a peristaltic vacuum pump **290** that communicates with the vitrectomy cutter **216** through a vacuum line **298**. As illustrated in FIG. **7**, in some embodiments, the peristaltic vacuum pump **290** may be provided in the housing **202**, such as on a housing side panel **206**. A stepper motor driver **296** located within the housing **202** may functionally interface with the peristaltic vacuum pump **290**. The microprocessor **236** may controllably interface with the stepper motor driver **296**. By operation of the peristaltic vacuum pump **290**, responsive to actuation by the stepper motor driver **296** via input from the microprocessor **236**, the peristaltic vacuum pump **290** may apply vacuum pressure to the vitrectomy cutter **216** through the vacuum line **298**.

(37) As further illustrated in FIG. **8**, a vacuum sensor **294** located within the housing **202** may interface with the vacuum line **298** through a vacuum sensor connector **292**. As shown in FIG. **7**, the vacuum sensor connector **292** can be arranged on a housing side panel **206** or other panel of the housing **202**, and preferably on the same housing side panel **206** as the peristaltic vacuum pump **290**. The microprocessor **236** may communicate with the vacuum sensor **294** through a feedback line **276**, shown in FIG. **8**. The vacuum sensor **294** may monitor the vacuum pressure in the vacuum line **298** through the vacuum sensor connector **292**, and may transmit vacuum pressure feedback which corresponds to the measured vacuum pressure to the microprocessor **236**. Responsive to the vacuum pressure feedback, the microprocessor **236** may modulate the operational speed of the peristaltic vacuum pump **290**, and hence, the vacuum pressure in the vacuum line **298** through the stepper motor driver **296**.

(38) As illustrated in FIG. **6**, in some embodiments, a pinch valve **278** may be provided on the housing **202**, such as on a housing side panel **206** opposite to the housing side panel **206** on which the peristaltic vacuum pump **290** is located. A flexible pinch valve tube **274**, which may be made from silicone, for example and without limitation, may extend through the pinch valve **278**. The pinch valve **278** may normally be disposed in a closed configuration in which the pinch valve **278** closes or pinches the pinch valve tube **274**. An irrigation fluid supply by gravity **275** may communicate with an inlet end of the pinch valve tube **274**. The pinch valve tube **274** may discharge into the eye through a sleeve alongside the cutting blade **220** of the vitrectomy cutter **216** or another separate channel or infusion line (as shown in FIG. **8**), as known in the art. A variable control foot pedal **240** (FIG. **8**) may operationally interface with the pinch valve **278**. Accordingly, responsive to depression of the variable control foot pedal **240**, the pinch valve **278** may open and release the pinch valve tube **274** so that fluid is allowed to flow through the pinch valve tube **274**. Consequently, irrigation fluid may flow from the irrigation fluid supply by gravity **275** through the pinch valve tube **274** and discharge through the sleeve parallel to the cutting blade **220** or another separate channel or infusion line **279**, as shown, to irrigate the eye **170** to maintain pressure in the eye **170** and prevent collapse.

(39) As further illustrated in FIG. **6**, a pinch valve button **280** may be provided on the housing **202**, such as on the same housing side panel **206** as the pinch valve **278**. The pinch valve button **280** may operably interface with the pinch valve **278** in such a manner that upon first depression of the pinch valve **278**, the pinch valve button **280** is permanently open for continuous irrigation of the

eye **170** through the pinch valve tube **274** such that the operator need not continually depress the variable control foot pedal **240**. Upon second depression of the pinch valve **278**, the pinch valve **278** may permanently close to prevent irrigation of the eye **170**.

(40) As illustrated in FIGS. **5** and **8**, a vacuum limit dial **282** on the housing **202** may interface with the microprocessor **236**. The vacuum limit dial **282** may facilitate selection of the maximum value, magnitude or level of vacuum pressure which the peristaltic vacuum pump **290** (FIG. **8**) applies to the vitrectomy cutter **216** through the vacuum line **298**. The vacuum limit dial **282** may have various settings (such as, but not limited to, four maximum vacuum value settings).

(41) As further illustrated in FIG. **6**, a volume control dial **284** on the housing **202** may interface with the microprocessor **236** (FIG. **8**). The volume control dial **284** may include an internal tone generator which indicates the various values, magnitudes or levels of vacuum pressure which the peristaltic pump **290** applies to the vitrectomy cutter **216**. The volume control dial **284** may be rotated to adjust the volume level of the internal tone generator.

(42) A flow rate selector dial **286** on the housing **202** may interface with the microprocessor **236**. The flow rate selector dial **286** may be rotated to select the operational pumping rate of the peristaltic vacuum pump **290** and may have multiple settings ranging from slow to fast.

(43) A mode selector dial **288** on the housing **202** may interface with the microprocessor **236**. The mode selector dial **288** may be rotated to select among multiple modes of operation of the anterior vitrectomy apparatus **200**. The modes of operation may include: (1) a first mode, in which operation is completely manual (no vacuum pump or irrigation); (2) a second mode, in which operation takes place with vacuum and irrigation only, without cutting; (3) a third mode, in which irrigation occurs first, cutting by the handpiece **218** takes place secondly, and vacuuming by the handpiece **218** takes place thirdly, depending on the degree or depth to which the variable control foot pedal **240** is pressed (e.g., initial pedal engagement starts irrigation, then the first half depression of the variable control foot pedal **240** engages the cutting action of the cutting blade **220**, and the second half depression of the variable control foot pedal **240** engages vacuuming); and (4) a fourth mode, in which irrigation occurs first, then the handpiece **218** vacuums second, and then the handpiece **218** cuts third, depending on the degree or depth to which the variable control foot pedal **240** is pressed (e.g., initial pedal engagement starts irrigation, then the first half depression of the variable control foot pedal **240** engages vacuuming, and the second half depression of the variable control foot pedal **240** engages the cutting action of the cutting blade **220**). For the purpose of executing the aforementioned third and fourth modes, the variable control foot pedal **240** may include a momentary switch that activates irrigation upon slightly pressing the variable control foot pedal **240**, and a potentiometer which is engaged when further depression of the variable control foot pedal **240** occurs, in order to activate cutting and vacuuming in the order corresponding to the specific mode and in dependence of the degree of compression of the variable control foot pedal **240**.

(44) Application of the anterior vitrectomy apparatus **200** may be similar to that which was heretofore described with respect to the anterior vitrectomy apparatus **100** in FIGS. **1-5**. The peristaltic vacuum pump **290** (FIG. **8**) may be operated to vacuum the vitreous humor from the eye **170** (FIG. **5**) of the patient **168**. The pinch valve **278**, pinch valve button **280**, vacuum limit dial **282**, volume control dial **284**, flow rate selector dial **286** and mode selector dial **288** may be manipulated to control various operational parameters of the peristaltic vacuum pump **290**, typically as was heretofore described.

(45) The illustrations of FIGS. **9** and **10** show an anterior vitrectomy apparatus **300** in accordance with a third illustrative embodiment of the present invention. The anterior vitrectomy apparatus **300** is analogous to the anterior vitrectomy apparatus **200**, and further includes a micro air pump **310** in the housing **202**. The micro air pump **310** is provided in fluid communication with an air tubing connector **312** on the housing **202**, such as on the housing front panel **208**. An external air tubing **314** (FIG. **10**) may be connected to the air tubing connector **312** and may provide fluid

communication between the air tubing connector **312** and a line configured to be introduced into the eye. For instance, an air tubing **314** may connect the air tubing connector **312** to the same line that is providing fluid irrigation into the eye via the pinch valve **278** (e.g., through the sleeve parallel to the cutting blade **220** or another separate channel or infusion line **279**, as shown). The micro air pump **310** may generate pressurized air and pump the pressurized air, via the air tubing connector **312**, to the external air tubing **314** and into the eye.

(46) The anterior vitrectomy apparatus **300** further comprises an internal light source or LED **320** controlled by a LED driver **322**. The LED **320** provides a source of light at a light connector **324** on the housing **202**, such as on the housing front panel **208**. A fiber-optic cable **326** (FIG. **10**) may be coupled to or plugged into the light connector **324**. A fiber optic handpiece **328** or light pipe may be coupled to the fiber-optic cable **326** and may be inserted into the eye, illuminating the eye from the inside when the LED **320** is activated by the LED driver **322** during operation of the anterior vitrectomy apparatus **300**.

(47) The illustration of FIG. **11** shows an anterior vitrectomy apparatus **400** in accordance with a fourth illustrative embodiment of the present invention. The anterior vitrectomy apparatus **400** of FIG. **11** may be similar to the anterior vitrectomy apparatus **300** shown in FIGS. **9** and **10**. For clarity purposes, the same reference numerals are being used in FIG. **11** to indicate components which are similar or identical to those of the anterior vitrectomy apparatus **300** of FIGS. **9** and **10** and prior embodiments shown in prior figures. Similarly to previous embodiments, the anterior vitrectomy apparatus **400** comprises housing **202**, which is preferably shaped and sized to fit in a human hand.

(48) The anterior vitrectomy apparatus **400** is however different from the anterior vitrectomy apparatus **300** in several aspects. Firstly, the anterior vitrectomy apparatus **400** lacks an air compressor connector (such as air compressor connector **260** of FIG. **10**) and is not configured to be connected to an external air compressor (such as air compressor **248** of FIG. **10**). Secondly, the anterior vitrectomy apparatus **400** comprises an air pump or air compressor **410** housed inside or contained within the housing **202**. The air compressor **410** may be operatively connected to the microprocessor **236** via one or more electrical control lines, which have been depicted as a single broken line. In some embodiments, the air compressor **410** may be powered by an external power source such as that which powers the microprocessor **236** and other applicable components via the power input port **264** (FIG. **7**).

(49) The internal air compressor **248** is configured to provide compressed air to the regulator **250**, for the regulator **250** to direct to the pressure reservoir **232** and subsequent components as described heretofore with reference to previous embodiments. In some embodiments, such as the present embodiment, the air compressor **248** may provide compressed air to the regulator **250** via the release valve **412**. The release valve **412**, which may also be contained within the housing **202**, may be configured to release a volume of air outside the housing **202**, via an air exhaust **414**, in the event of an excessive pressure being sensed by the release valve **412** at the air compressor **410**. The release valve **412** may be operatively connected to and controlled by the microprocessor **236** via one or more electrical control lines, which have been depicted as a single broken line. In some embodiments, the release valve **412** may be powered by an external power source such as that which powers the microprocessor **236** and other applicable components via the power input port **264** (FIG. **7**).

(50) The anterior vitrectomy apparatus **400** may further include a pressure sensor **416**, configured to measure air pressure immediately after the air compressor **410** (or at an output thereof), i.e. between the air compressor **410** and the release valve **412**. The pressure sensor **416** may be operatively connected to the microprocessor **236** via one or more electrical control lines, which have been depicted as a single broken line. In some embodiments, the pressure sensor **416** may be powered by an external power source such as that which powers the microprocessor **236** and other applicable components via the power input port **264** (FIG. **7**).

(51) In some embodiments, operation of the anterior vitrectomy apparatus **400** with respect to the air compressor **410**, release valve **412**, and pressure sensor **416**, may be as follows. As was heretofore described with reference to the anterior vitrectomy apparatus **200**, a user may operate or engage the variable control foot pedal **240** (e.g., according to the third or fourth modes) to initiate cutting by the vitrectomy cutter **216**. When the variable control foot pedal **240** is not engaged, or is engaged in a manner not configured to initiate cutting, the microprocessor **236** may monitor the air pressure immediately after the air compressor **410** via the pressure sensor **416**; in the event that the air pressure is detected by the pressure sensor **416** to be greater than a predetermined threshold, the microprocessor **236** may responsively open the release valve **412** to cause compressed air to be exhausted via the air exhaust **414**. The predetermined threshold is greater than or equal to zero and indicates a slight air compression or no air compression, respectively, and more preferably, is about zero (i.e. about atmospheric air pressure or about zero air compression). In this way, the air compressor **410** is maintained with a low, and more preferably, zero pressure until subsequent engagement of the variable control foot pedal **240** by the user to initiate cutting; i.e. the air compressor **410** is thereby allowed to engage without air pressure resistance. When the variable control foot pedal **240** is engaged in a manner configured to initiate cutting, the microprocessor **236** activates the air compressor **410** and closes the release valve **412** such that air is not exhausted through the air exhaust **414**, allowing compressed air from the air compressor **410** to flow generally entirely towards the regulator **250**, pressure reservoir **232**, and so forth.

(52) The anterior vitrectomy apparatus **400** of the present embodiment may thus be operated without relying on an external gas source, thereby rendering the anterior vitrectomy apparatus **400** more self-contained and easier to carry, transport, store, and operate. Furthermore, with respect to conventional anterior vitrectomy apparatuses in which compressed air is provided by external compressed air canisters, the anterior vitrectomy apparatus **400** provides the additional advantage that it does not require external gas replacement or refilling, and is thus always readily available for use. The anterior vitrectomy apparatus **400** of the present embodiment generally incorporates all or most necessary features into a small and portable unit which may be completely independent of external gas sources.

(53) It should be noted that further embodiments are contemplated in which the anterior vitrectomy apparatus **100** and **200** of the present disclosure are provided with the internal air compressor **410**, release valve **412**, air exhaust **414** and pressure sensor **416**, and associated methods of operation (which may also involve the microprocessor **236**) as heretofore described with reference to the anterior vitrectomy apparatus **400**.

(54) It will be appreciated by those skilled in the art that the anterior vitrectomy apparatus **100**, **200**, **300**, **400** is capable of being advantageously used, for instance, in rural settings where cataract apparatus are often needed but not available. The anterior vitrectomy apparatus **100**, **200**, **300**, **400** can provide optimum performance specifications (cut rate of 2000+ cuts per minute, integrated peristaltic pump with linear vacuum control and irrigation control) while fitting in the palm of the hand.

(55) Since many modifications, variations, and changes in detail can be made to the described preferred embodiments of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Furthermore, it is understood that any of the features presented in the embodiments may be integrated into any of the other embodiments unless explicitly stated otherwise. The scope of the invention should be determined by the appended claims and their legal equivalents.

Claims

1. An anterior vitrectomy apparatus, comprising: an air compressor, configured to generate compressed air; a pressure reservoir, configured to store compressed air received from the air

compressor; a regulator, configured to vary a delivery of compressed air from the air compressor to the pressure reservoir to adjust air pressure within the pressure reservoir; a speed control switch; an air control valve, configured to receive compressed air from the pressure reservoir and to produce pulses of compressed air in accordance with a selective adjustment of the speed control switch; an air tubing connector, configured to receive said pulses of compressed air from the air control valve; an air tubing, connectable to the air tubing connector; a handheld vitrectomy cutter, connectable to the air tubing to receive said pulses of compressed air from the air control valve via the air tubing connector and the air tubing; a processor controllably interfacing with the speed control switch, the processor comprising a memory storing software instructions configured to cause the processor to execute the operation of: producing an operation of the air control valve through the speed control switch such that the air control valve delivers said pulses of compressed air to the air tubing connector to drive the handheld vitrectomy cutter; and a housing containing the air compressor, the pressure reservoir, the regulator, the air control valve, and the processor, wherein the housing is sized to fit in and be held by a human hand, and further wherein the air tubing connector and the speed control switch are arranged on an exterior of the housing.

2. The anterior vitrectomy apparatus of claim 1, further comprising a release valve, the release valve contained inside the housing and configured to selectively exhaust compressed air from between the compressor and the regulator, wherein the processor controllably interfaces with the release valve and is configured to cause the release valve to exhaust compressed air from between the air compressor and the regulator in the absence of said operation of the air control valve by the processor.

3. The anterior vitrectomy apparatus of claim 2, further comprising a pressure sensor, the pressure sensor contained inside the housing and configured to sense air pressure between the air compressor and the regulator, wherein the processor controllably interfaces with the pressure sensor and is configured to cause the release valve to exhaust compressed air from between the air compressor and the regulator when the processor detects a pressure sensing by the pressure sensor between the air compressor and the regulator in the absence of said operation of the air control valve by the processor, the pressure sensing greater than a predetermined threshold.

4. The anterior vitrectomy apparatus of claim 3, wherein the predetermined threshold is about zero.

5. The anterior vitrectomy apparatus of claim 2, wherein the processor is configured to cause the release valve not to exhaust compressed air from between the air compressor and the regulator during said operation of the air control valve by the processor.

6. The anterior vitrectomy apparatus of claim 1, further comprising a pressure gauge inside the housing, the pressure gauge interfacing with the processor and the pressure reservoir and configured to indicate pressure of air in the pressure reservoir to the processor.

7. The anterior vitrectomy apparatus of claim 1, further comprising a syringe connectable to the handheld vitrectomy cutter to suction and collect vitreous humor from the handheld vitrectomy cutter.

8. The anterior vitrectomy apparatus of claim 1, further comprising a foot pedal connector on the exterior of the housing, the foot pedal connector configured for the coupling thereto of a foot pedal external to the housing such that the foot pedal connector connects the foot pedal to the processor to facilitate said operation of the air control valve.

9. The anterior vitrectomy apparatus of claim 1, wherein the speed control switch comprises a rotatable dial.

10. The anterior vitrectomy apparatus of claim 1, further comprising a reset button on housing, the reset button configured to facilitate a resetting of the processor.

11. The anterior vitrectomy apparatus of claim 1, further comprising a peristaltic pump and a vacuum line, wherein the peristaltic pump is located inside the housing and the vacuum line provides fluid communication between the peristaltic pump and the handheld vitrectomy cutter to suction and collect vitreous humor from the handheld vitrectomy cutter.

12. The anterior vitrectomy apparatus of claim 11, wherein the peristaltic pump is driven by a stepper motor driver located inside the housing and interfacing with the processor.
13. The anterior vitrectomy apparatus of claim 11, further comprising a vacuum sensor in fluid communication with the vacuum line and configured to provide a measurement of air vacuum pressure in the vacuum line to the processor.
14. The anterior vitrectomy apparatus of claim 1, further comprising an air pump and a second air tubing, wherein the air pump is located inside the housing and is configured to provide pressurized air, and the air tubing provides fluid communication between the second air pump and an infusion line external to the housing and configured to be inserted into a subject's eye.
15. The anterior vitrectomy apparatus of claim 1, further comprising a pinch valve, a pinch valve tube and an infusion line, wherein the pinch valve is located inside the housing and the infusion line is configured to be inserted into a subject's eye, and further wherein the pinch valve tube provides fluid communication between a source of irrigation fluid and the infusion line through the pinch valve.
16. The anterior vitrectomy apparatus of claim 15, wherein the source of irrigation fluid is configured to feed irrigation fluid to the pinch valve by gravity.
17. The anterior vitrectomy apparatus of claim 15, further comprising an air pump and a second air tubing, wherein the air pump is located inside the housing and is configured to provide pressurized air, and the air tubing provides fluid communication between the second air pump and the infusion line.
18. The anterior vitrectomy apparatus of claim 1, further comprising a light source and a fiber-optic cable, wherein the light source is arranged inside the housing and the fiber-optic cable is configured to transport light emitted by the light source to a fiber-optic handpiece external to the housing and configured to be inserted into a subject's eye.
19. An anterior vitrectomy apparatus, comprising: an air compressor, configured to generate compressed air; a pressure reservoir, configured to store compressed air received from the air compressor; a regulator, configured to vary a delivery of compressed air from the air compressor to the pressure reservoir to adjust air pressure within the pressure reservoir; a release valve, configured to selectively exhaust compressed air from between the compressor and the regulator; a pressure sensor, configured to sense air pressure between the air compressor and the regulator; a speed control switch; an air control valve, configured to receive compressed air from the pressure reservoir and to produce pulses of compressed air in accordance with a selective adjustment of the speed control switch; an air tubing connector, configured to receive said pulses of compressed air from the air control valve; an air tubing, connectable to the air tubing connector; a handheld vitrectomy cutter, connectable to the air tubing to receive said pulses of compressed air from the air control valve via the air tubing connector and the air tubing; a processor controllably interfacing with the speed control switch, the processor comprising a memory storing software instructions configured to cause the processor to execute the operation of: producing an operation of the air control valve through the speed control switch such that the air control valve delivers said pulses of compressed air to the air tubing connector to drive the handheld vitrectomy cutter, causing the release valve to exhaust compressed air from between the air compressor and the regulator when the processor detects a pressure sensing by the pressure sensor between the air compressor and the regulator in the absence of said operation of the air control valve, the pressure sensing greater than a predetermined threshold, and causing the release valve not to exhaust compressed air from between the air compressor and the regulator during said operation of the air control valve; and a housing containing the air compressor, the pressure reservoir, the regulator, the release valve, the pressure sensor, the air control valve, and the processor, wherein the housing is sized to fit in and be held by a human hand, and further wherein the air tubing connector and the speed control switch are arranged on an exterior of the housing.
20. An anterior vitrectomy apparatus, comprising: an air compressor, configured to generate

compressed air; a pressure reservoir, configured to store compressed air received from the air compressor; a regulator, configured to vary a delivery of compressed air from the air compressor to the pressure reservoir to adjust air pressure within the pressure reservoir; a release valve, configured to selectively exhaust compressed air from between the compressor and the regulator; a pressure sensor, configured to sense air pressure between the air compressor and the regulator, a speed control switch; an air control valve, configured to receive compressed air from the pressure reservoir and to produce pulses of compressed air in accordance with a selective adjustment of the speed control switch; an air tubing connector, configured to receive said pulses of compressed air from the air control valve; an air tubing, connectable to the air tubing connector; a handheld vitrectomy cutter, connectable to the air tubing to receive said pulses of compressed air from the air control valve via the air tubing connector and the air tubing; a processor controllably interfacing with the speed control switch, the processor comprising a memory storing software instructions configured to cause the processor to execute the operation of: producing an operation of the air control valve through the speed control switch such that the air control valve delivers said pulses of compressed air to the air tubing connector to drive the handheld vitrectomy cutter, causing the release valve to exhaust compressed air from between the air compressor and the regulator when the processor detects a pressure sensing by the pressure sensor between the air compressor and the regulator in the absence of said operation of the air control valve, the pressure sensing greater than a predetermined threshold of about zero, and causing the release valve not to exhaust compressed air from between the air compressor and the regulator during said operation of the air control valve; and a housing containing the air compressor, the pressure reservoir, the regulator, the release valve, the pressure sensor, the air control valve, and the processor, wherein the housing is sized to fit in and be held by a human hand, and further wherein the air tubing connector and the speed control switch are arranged on an exterior of the housing.
